

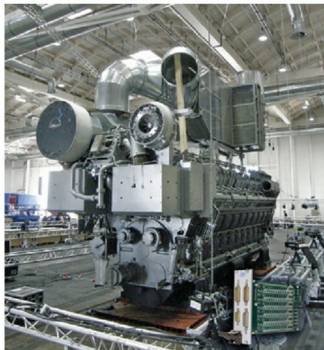


Fig. 4: A shipboard diesel engine with many temperature sensors to monitor its operation. Using a PXI chassis, these sensors are simulated when testing the engine control

In addition to the precision resistor modules, switch cards for both signal and high-power switching, isolated power supply cards and MIL-STD-1553 bus analyser modules were used. The main functions of the switching modules were to physically isolate interface paths, simulate RF switch feedback contacts and distribute load channels across multiple paths.

### 3 Diesel engine temperature simulation

**Customer.** A manufacturer of large diesel engines, whose products are used in ships and power stations. Some of these diesel engines are being manufactured for a new class of container ships, which are among the world's largest cargo carriers.

**Application.** The controllers for the massive diesel engines in ocean-going vessels require many temperature sensors in order to determine whether the engine is operating correctly. This is primarily because of the sheer size of the engine block. Many sensors are required to ensure that the engine operates in the correct temperature range

and hot spots are identified quickly. It would be difficult to have an engine block available for system test, not to mention the time it would take for the engine to reach operating temperature. Sensor simulation provides a cost-effective means of testing the controller without having an engine on hand. This requirement involved the support of ECU development for 40,000kW ship diesel engines. In terms of horsepower (HP), 40,000 kW equates to approximately 55,000 HP. Besides other ECU tasks, it was required to simultaneously simulate 144 channels of PT100 temperature sensors. These sensors are resistive temperature devices (RTDs) as opposed to thermocouples. The temperature range needed here was  $-20^{\circ}\text{C}$  to  $+250^{\circ}\text{C}$ , with an equivalent resistance range of 92.160342 ohms to 194.074250 ohms. Extreme precision was also required, demanding accuracies within  $0.11^{\circ}\text{C}$  at  $-20^{\circ}\text{C}$  (equivalent resistance deviation within 43 milli-ohms) and  $0.65^{\circ}\text{C}$  at  $250^{\circ}\text{C}$  (equivalent resistance deviation within 220 milli-ohms).

The customer's original method of sensor simulation was achieved manually. This comprised 288 precision potentiometers (covering both coarse and fine adjustment) along with 144 switches for short circuits, and 144 switches for open circuits in the simulation of faulty wiring connections to a sensor. All these components were adjusted and controlled by hand. The need for automation was obvious in order to save time and improve performance and repeatability.

**Solution.** With various measurement and stimulus modules already available in PXI and a requirement to support RTOS software, PXI was seen as the natural choice of platform for new product design.

The solution developed was a 3U PXI module that supports either six channels of RTD simulation (in one slot) or 18 channels (in two slots). Based upon existing design principles, the module provides a setting resolution of 2 milli-ohms and a resistance accuracy of better than 0.1 per cent on all channels. Each simulation channel is able to provide a short or open-circuit setting in order to simulate faulty wiring connections to a sensor. Furthermore, the calibration or verification of each resistor channel is possible by connecting the module calibration port to a high-performance digital multimeter.

The use of simple resistance value calls makes programming simple. This was achieved by using an application programming interface (API) to convert temperature request into resistance request through a model of the sensor used in the real engine.

The end product enabled a cost-effective method of simulating either PT100 or PT1000 RTDs with very high accuracy. Eight modules achieved the 144 channels of PT100 required in this application (18-channel two-slot modules), accommodated in sixteen slots of 3U PXI chassis space.

### A 'simulated' conclusion

From these three applications it is clear that PXI products can simulate the signals necessary for a variety of applications including consumer, transportation and aerospace. PXI can achieve the densities required to make the system manageable on the test floor—accurate enough to meet customer demands and support specialised environments like RTOS. ●

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